U24CSB462

Natural Language Processing and Artificial Intelligence

COURSE OBJECTIVES:

1. Understand the foundational concepts of Natural Language Processing (NLP) and Artificial Intelligence (AI).
2. Learn the architecture and techniques behind language modeling, parsing, tagging, and machine translation.
3. Apply machine learning and deep learning models for NLP tasks.
4. Gain insight into real-world AI applications using NLP.
5. Analyze the ethical and societal impacts of AI and NLP systems.

COURSE OUTCOMES:

CO1: Understand the basic principles and techniques of AI and NLP.
CO2: Apply parsing and language modeling algorithms to solve NLP tasks.
CO3: Use supervised and unsupervised learning models in NLP.
CO4: Design and evaluate AI models for sentiment analysis, summarization, etc.
CO5: Critically assess AI system design with ethical and practical implications.

Unit -I Introduction to AI and NLP

Overview of Artificial Intelligence, History and evolution of NLP, Natural Language Understanding vs. Generation, Applications of NLP

Unit -II NLP Fundamentals

Text preprocessing (tokenization, stemming, lemmatization), POS tagging Syntax and semantics, Language modeling (N-gram, probabilistic models)

Unit -III Parsing and Tagging

Shallow parsing, Dependency and constituency parsing, Named Entity Recognition (NER), Grammar-based and statistical parsing

Unit -IV Machine Learning for NLP

Word Embeddings: Word2Vec, GloVe, Supervised models: Naive Bayes, SVM, Decision Trees,
Unsupervised models: Clustering, Topic Modeling (LDA), Sequence models: HMM, CRF

Unit -V Deep Learning and NLP Applications

RNN, LSTM, GRU, Transformers and BERT, Applications: Machine Translation, Chatbots, Sentiment Analysis, Ethics in NLP and Bias in AI.

TEXT BOOKS:

1. Daniel Jurafsky & James H. Martin – Speech and Language Processing, 3rd Edition, Pearson.
2. Stuart Russell & Peter Norvig – Artificial Intelligence: A Modern Approach, 4th Edition, Pearson.

REFERENCE BOOKS:

1. Steven Bird, Ewan Klein, and Edward Loper – Natural Language Processing with Python, O'Reilly Media.
2. Elaine Rich, Kevin Knight – Artificial Intelligence, Tata McGraw-Hill.
3. Yoav Goldberg – Neural Network Methods in NLP, Morgan & Claypool.

4. Chris Manning & Hinrich Schütze – Foundations of Statistical NLP, MIT Press
CASE STUDIES:
1. Sentiment Analysis on Social Media Data
2. Chatbot for Student Query Resolution
NLP and Artificial Intelligence – Lab Experiments
1. Introduction to Python for AI and NLP
Write basic Python programs for data types, control structures, and functions.
2. Text Preprocessing Techniques
Implement tokenization, stop-word removal, stemming, and lemmatization.
3. Word Frequency and Vocabulary Analysis
Perform word count and frequency distribution on a given text corpus.
4. N-grams and Language Modeling
Generate unigrams, bigrams, and trigrams from a text dataset.
5. Text Vectorization Techniques
Implement Bag of Words (BoW) and TF-IDF representations.
6. Part-of-Speech (POS) Tagging
Apply POS tagging using NLP libraries.
7. Named Entity Recognition (NER)
Identify named entities such as person, location, and organization.
8. Text Classification using Machine Learning
Build a simple text classifier (spam detection or sentiment analysis).
9. Sentiment Analysis
Analyze sentiment polarity of user reviews or social media text.
10. Rule-Based Chatbot
Design a simple rule-based chatbot using pattern matching.
11. Introduction to Machine Learning for AI
Implement a basic supervised learning algorithm (Linear Regression / Naive Bayes).
12. Search Algorithms in AI
Implement BFS or DFS for problem solving.
13. Heuristic-Based Search
Implement A* or Best-First Search algorithm.
14. Basic Recommendation System
Develop a simple content-based recommendation system.
30 Hours
15. Mini Project – NLP/AI Application
Develop a basic NLP or AI application such as chatbot, text summarizer, or sentiment Analyzer

UNIT I

THEORY TOPICS NOTES

Introduction to NLP and AI

1. What is Artificial Intelligence (AI)?

Definition:

- Artificial Intelligence (AI) is the **branch of computer science that creates intelligent machines capable of performing tasks that normally require human intelligence**.
- Examples: Speech recognition, self-driving cars, recommendation systems, chatbots.

Key Points to Teach:

- AI aims to **simulate human intelligence** in machines.
- AI can **learn, reason, and self-correct**.

Types of AI:

1. **Narrow AI (Weak AI):** Performs specific tasks. Example: Siri, Google Search.
2. **General AI (Strong AI):** Human-level intelligence across multiple tasks. (Still theoretical)
3. **Super AI:** Surpasses human intelligence (hypothetical/future).

2. What is Natural Language Processing (NLP)?

Definition:

- NLP is a **field of AI** that enables computers to **understand, interpret, and generate human language**.

Key Points to Teach:

- NLP bridges the gap between **human communication and computer understanding**.
- It is **multidisciplinary**: AI + linguistics + computer science.

Applications of NLP:

- Chatbots (Customer service)
- Machine Translation (Google Translate)
- Sentiment Analysis (Social media reviews)
- Text Summarization
- Spam Detection (Email filtering)

3. NLP vs NLU vs NLG

Term	Full Form	Description	Example
NLP	Natural Language Processing	Overall processing of human language	Tokenizing, POS tagging
NLU	Natural Language Understanding	Understanding meaning from text	Sentiment analysis, question answering

Term	Full Form	Description	Example
NLG	Natural Language Generation	Generating human-like language from data	Chatbot responses, automated reports

Use a simple analogy:

- **NLU = Reading & Understanding**
- **NLG = Writing / Speaking**

4. Importance of NLP

Why NLP is important:

- **Automates communication:** Chatbots, email responses.
- **Analyzes huge data:** Social media sentiment, customer reviews.
- **Enhances human-computer interaction:** Voice assistants, smart devices.
- **Supports AI applications:** Search engines, recommendation systems.

Examples to show:

- "Alexa, play music" → Voice command understood and action executed.
- Social media sentiment: Companies know public opinion automatically.

5. Phases of NLP

1. Text Preprocessing:

- Clean and normalize text for analysis.
- Steps: Tokenization, Stopword removal, Stemming, Lemmatization, Lowercasing.

2. Language Modeling:

- Understanding probability of word sequences.
- N-grams, probabilistic models.

3. Parsing & Tagging:

- Identify grammatical structures and parts-of-speech.
- Shallow parsing, Dependency parsing, NER (Named Entity Recognition).

4. Semantic Analysis:

- Understanding the meaning of text.
- Word sense disambiguation, context understanding.

5. Application Layer:

- Using processed text for AI tasks: Sentiment analysis, summarization, machine translation, chatbots.

Visual Aid Idea: Draw a pipeline showing:

Raw Text → Preprocessing → Language Modeling → Parsing → Semantic Analysis → Applications

6. NLP Tools and Libraries

- **NLTK (Natural Language Toolkit):** Python library for text processing.
- **SpaCy:** Industrial-strength NLP library, fast and efficient.
- **Gensim:** Topic modeling and document similarity analysis.
- **TextBlob:** Simplified API for sentiment analysis and basic NLP.
- **Hugging Face Transformers:** Pre-trained models like BERT, GPT for advanced NLP tasks.

7. Trends & Future of NLP

- **Pre-trained language models:** BERT, GPT, T5, etc.

- **Conversational AI / Chatbots:** Human-like interactions.
- **AI in education, healthcare, and business:** Automated reports, diagnostics, sentiment analysis.
- **Multilingual NLP:** Translation between multiple languages in real-time.
- **Ethics and Bias:** Ensuring fairness and avoiding algorithmic discrimination.

History and Evolution of Natural Language Processing (NLP)

- Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language. The evolution of NLP has passed through several important stages.

1. Early Beginnings (1950s–1960s)

- The concept of NLP began in the 1950s with the famous **Turing Test**, proposed by Alan Turing to evaluate machine intelligence. Early NLP systems were mainly focused on **machine translation** using simple dictionary-based approaches.

2. Rule-Based Era (1970s–1980s)

- In this period, NLP systems were developed using **handwritten grammatical rules and linguistic patterns**. These systems required human experts to define rules for sentence structure, syntax, and meaning. Although accurate for limited domains, they lacked scalability.

3. Statistical NLP Era (1990s)

- With the availability of large digital text corpora, NLP shifted towards **probabilistic and statistical models**. Techniques like **Hidden Markov Models (HMM)** and **N-gram models** were introduced to handle speech recognition and tagging tasks.

4. Machine Learning Era (2000s)

- Machine learning techniques such as **Naïve Bayes, Support Vector Machines (SVM), and Decision Trees** were applied to NLP tasks like text classification and sentiment analysis. Systems began learning patterns automatically from data.

5. Deep Learning Era (2010–Present)

- The introduction of **Neural Networks, RNN, LSTM, and Transformers** revolutionized NLP. Pre-trained models like **BERT, GPT, and T5** enabled machines to understand context and generate human-like responses.

6. Modern NLP

- Today, NLP is widely used in chatbots, voice assistants, translation systems, and recommendation engines, making human-computer interaction more natural and intelligent.

NLP Vs NLU vs NLG

Aspect	NLP (Natural Language Processing)	NLU (Natural Language Understanding)	NLG (Natural Language Generation)
Definition	Overall field of AI that deals with **processing human language**	Subfield of NLP focused on **understanding the meaning of text**	Subfield of NLP focused on **generating human-like text or speech**

Aspect	NLP (Natural Language Processing)	NLU (Natural Language Understanding)	NLG (Natural Language Generation)
Purpose	To enable machines to **read, analyze, and process text**	To **interpret text semantically and contextually**	To **create coherent text from structured data or models**
Focus	Syntax + Semantics + Context	Semantics + Intent + Context	Syntax + Grammar + Coherence
Input	Raw text or speech	Text that needs understanding	Structured data or intermediate representation
Output	Processed language data, tokenization, POS tags, embeddings	Understanding of meaning, sentiment, intent, entities	Human-readable text or responses
Techniques	Tokenization, POS tagging, Lemmatization, Stemming, Parsing	Sentiment analysis, NER (Named Entity Recognition), Question answering, Intent detection	Text summarization, Chatbot response generation, Machine translation, Report generation
Example Use Case	Tokenize a paragraph, POS tagging of a sentence	Chatbot understanding "Book a flight to Paris tomorrow"	Chatbot responding: "Your flight to Paris is booked for tomorrow"
Goal	Bridge human language and machine processing	Interpret human language correctly	Communicate machine decisions back to humans naturally
Key Libraries/Tools	NLTK, SpaCy, Gensim, TextBlob	NLTK, SpaCy, BERT, Hugging Face Transformers	Hugging Face Transformers, GPT, T5, OpenAI API
Analogy	Reading a book	Understanding what the book means	Writing a summary or explaining the book

Applications and Importance of NLP

1. What is NLP? ***(Quick Recap)***

- **Natural Language Processing (NLP)** is a **branch of AI** that enables machines to **understand, interpret, and generate human language**.
- It combines **computer science, linguistics, and AI** to bridge the gap between humans and machines.

2. Applications of NLP

NLP has **widespread applications** across industries. Here's a breakdown:

Application Area	Description	Example
Chatbots & Virtual Assistants	Automates customer service by understanding queries and responding	Siri, Alexa, Google Assistant
Machine Translation	Converts text or speech from one language to another	Google Translate, Microsoft Translator
Sentiment Analysis	Determines the emotion or opinion expressed in text	Social media monitoring, product reviews analysis
Text Summarization	Produces a concise summary from a large document	Automatic news summarization, legal document summary
Spam Detection	Filters unwanted emails by analyzing text content	Gmail spam filter
Speech Recognition	Converts spoken words into text	Voice-to-text apps, transcription services
Recommendation Systems	Understands user preferences from text reviews	Netflix, Amazon, Spotify recommendations
Named Entity Recognition (NER)	Identifies entities like names, dates, organizations	Extracting key information from resumes, news articles
Information Retrieval	Retrieves relevant information from large datasets	Search engines like Google
Healthcare Applications	Extracts useful insights from medical records	Patient data analysis, clinical decision support

3. Importance of NLP

Why NLP is crucial in modern AI systems:

1. **Automates Human Communication**
   - Enables machines to interact with humans naturally via text and speech.
   - Example: Customer support chatbots answer queries instantly.
2. **Handles Big Data Efficiently**
   - Extracts insights from large volumes of unstructured text.
   - Example: Social media monitoring to gauge public sentiment.
3. **Enhances Decision Making**
   - Provides actionable insights from textual data.
   - Example: Summarizing market reports for business strategy.
4. **Improves Human-Computer Interaction**
   - Makes digital assistants, apps, and devices more intuitive.
   - Example: Voice commands in smart homes.
5. **Enables Multilingual Communication**
   - Breaks language barriers via machine translation.
   - Example: Translating websites and apps for global users.
6. **Supports AI Applications**

    - Forms the backbone for AI models in NLP tasks.
    - Example: Sentiment analysis, chatbots, recommendation systems.

Phases of Natural Language Processing (NLP)

NLP is a **multi-step process** that transforms raw human language into a format machines can understand and use for intelligent applications. Each **phase focuses on a specific task** in the NLP pipeline.

1. Text Preprocessing

Purpose:

- Prepare raw text for analysis by cleaning and structuring it.

Key Steps:

1. **Tokenization:**
    - Splits text into **words, sentences, or phrases**.
    - Example: "I love NLP" → ["I", "love", "NLP"]
2. **Stopword Removal:**
    - Removes common words that **don't carry significant meaning**.
    - Example: "I love NLP" → ["love", "NLP"]
3. **Stemming:**
    - Reduces words to their **root form**.
    - Example: "running", "runs" → "run"
4. **Lemmatization:**
    - Converts words to their **dictionary form** using context.
    - Example: "better" → "good"
5. **Lowercasing & Cleaning:**
    - Convert all text to lowercase and remove punctuation, numbers, or special characters.

Tools: NLTK, SpaCy, TextBlob

2. Language Modeling

Purpose:

- Predict **the likelihood of word sequences** and understand text patterns.

Key Concepts:

1. **N-grams:**
    - Probabilistic models for sequences of words.
    - Example: Bigrams → "I love", "love NLP"
2. **Probabilistic Models:**
    - Estimate the **probability of the next word** given previous words.

Applications:

- Text prediction, autocomplete, spell check.

3. Parsing & Tagging

Purpose:

- Analyze **grammatical structure** and label words.

Key Steps:

1. **Part-of-Speech (POS) Tagging:**
    - Assigns grammatical labels like noun, verb, adjective.
    - Example: "I/PRON love/VERB NLP/NOUN"
2. **Shallow Parsing / Chunking:**
    - Groups words into **phrases** like noun phrases (NP).
3. **Dependency Parsing:**
    - Understands **relationships between words**.
    - Example: "I → love → NLP"
4. **Named Entity Recognition (NER):**
    - Identifies **entities**: names, dates, locations.
    - Example: "Barack Obama → PERSON, Paris → LOCATION"

Tools: NLTK, SpaCy

4. Semantic Analysis

Purpose:

- Understand the **meaning and context** of text.

Key Concepts:

1. **Word Sense Disambiguation:**
    - Determines the correct meaning of a word in context.
    - Example: "bank" → river bank or financial bank?
2. **Sentiment Analysis:**
    - Determines **emotions or opinions** in text.
3. **Coreference Resolution:**
    - Identifies when **different words refer to the same entity**.
    - Example: "John went home. He was tired." → He = John

5. Application / Task Layer

Purpose:

- Use processed and analyzed text for real-world NLP tasks.

Examples of Applications:

- **Chatbots:** Generate responses based on user input.
- **Machine Translation:** Translate text from one language to another.
- **Text Summarization:** Condense large documents into key points.
- **Recommendation Systems:** Suggest products based on text reviews.

NLP Trends and the Future of Natural Language Processing

Natural Language Processing (NLP) is one of the **fastest-growing areas in AI**, transforming how humans interact with machines. Understanding **current trends and future directions** helps students see its real-world impact and career potential.

1. Rise of Pre-Trained Language Models

Definition:

- Pre-trained models are **AI models trained on massive text datasets** to understand language patterns and context.

Examples:

- **BERT (Bidirectional Encoder Representations from Transformers)** – understands context both ways.
- **GPT (Generative Pre-trained Transformer)** – generates human-like text.
- **T5, RoBERTa, XLNet** – specialized models for summarization, translation, and question answering.

Impact:

- Reduces the need for training models from scratch.
- Improves **accuracy and efficiency** in NLP tasks.

2. Conversational AI and Chatbots

Trend:

- More **human-like virtual assistants** are appearing in customer support, healthcare, and education.

Examples:

- Chatbots for online shopping (Amazon), student queries (University portals).
- AI assistants like Siri, Alexa, Google Assistant.

Future Direction:

- Chatbots that can **understand emotions, context, and follow-up questions** naturally.

3. Multilingual NLP

Trend:

- NLP systems are increasingly able to **understand multiple languages**, breaking communication barriers.

Examples:

- Google Translate, Microsoft Translator, multilingual chatbots.

Future Direction:

- Real-time **speech-to-speech translation** in any language.
- Cross-lingual models capable of learning from one language and applying it to another.

4. Sentiment & Emotion Analysis

Trend:

- NLP models are improving in **detecting sentiment, opinion, and emotions** in text.

Examples:

- Social media monitoring for brand reputation.
- Customer feedback analysis for products and services.

Future Direction:

- Detect **nuanced emotions**, sarcasm, and context-specific sentiment.

5. Text Summarization and Knowledge Extraction

Trend:

- Automated summarization of large text data is in high demand.

Examples:

- News summarization, medical reports, legal document summarization.

Future Direction:

- Real-time summarization for **streaming data**.

- Extracting **actionable insights** from unstructured text in enterprises.

6. Ethics and Bias in NLP

Trend:

- Awareness is growing about **bias in NLP models** due to biased training data.

Examples:

- Gender or racial bias in text prediction or sentiment analysis.

Future Direction:

- Development of **fair, explainable, and transparent NLP models**.
- AI systems that **mitigate discrimination and bias**.

7. Integration with AI and IoT

Trend:

- NLP is being integrated with **AI systems and IoT devices** for smart applications.

Examples:

- Smart homes with voice assistants controlling appliances.
- AI-powered healthcare assistants providing insights from patient data.

Future Direction:

- Seamless interaction between humans, AI, and connected devices.
- Enhanced personalization and automation.

NLP Libraries in Python

Python has become the **most popular language for NLP** due to its rich ecosystem of libraries that simplify **text processing, analysis, and AI applications**.

1. NLTK (Natural Language Toolkit)

Overview:

- One of the **earliest and most widely used NLP libraries** in Python.
- Provides **tools for text preprocessing, tokenization, stemming, tagging, parsing, and more**.

Key Features:

- Tokenization: Split text into words or sentences.
- Stopwords: Remove common words like “the”, “is”, “and”.
- Stemming & Lemmatization: Reduce words to root forms.
- POS Tagging: Identify nouns, verbs, adjectives, etc.
- Corpora: Preloaded datasets for NLP tasks.

Example:

```
import nltk
from nltk.tokenize import word_tokenize
nltk.download('punkt')

text = "I love Natural Language Processing!"
tokens = word_tokenize(text)
print(tokens)
```

Output:

['I', 'love', 'Natural', 'Language', 'Processing', '!']

2. SpaCy

Overview:

- Industrial-strength NLP library, faster and more efficient than NLTK.
- Ideal for **production-level NLP applications**.

Key Features:

- Tokenization, Lemmatization, POS Tagging
- Named Entity Recognition (NER)
- Dependency Parsing
- Pre-trained models for multiple languages

Example:

```
import spacy

nlp = spacy.load("en_core_web_sm")
doc = nlp("Apple is looking at buying U.K. startup for $1 billion")

for ent in doc.ents:
    print(ent.text, ent.label_)
```

Output:

Apple ORG
U.K. GPE
$1 billion MONEY

3. Gensim

Overview:

- Library focused on **topic modeling, document similarity, and vector space modeling**.
- Works well with **large text corpora**.

Key Features:

- Word2Vec embeddings
- LDA (Latent Dirichlet Allocation) for topic modeling
- Doc2Vec for document embeddings

Example:

```
from gensim.models import Word2Vec

sentences = [["I", "love", "NLP"], ["Word embeddings are fun"]]
model = Word2Vec(sentences, vector_size=50, window=2, min_count=1, workers=4)
print(model.wv['NLP'])
```

Output:

- Vector representation of the word NLP (numerical embedding).

4. TextBlob

Overview:

- Simplified NLP library built on NLTK and Pattern.
- Great for **quick prototyping and sentiment analysis**.

Key Features:

- Sentiment Analysis (polarity & subjectivity)
- POS Tagging
- Noun Phrase Extraction
- Translation & Language Detection

Example:

```
from textblob import TextBlob

text = TextBlob("I love Natural Language Processing!")
print(text.sentiment)
```

Output:

```
Sentiment(polarity=0.5, subjectivity=0.6)
```

5. Hugging Face Transformers

Overview:

- Modern library for **state-of-the-art NLP** using **deep learning models**.
- Supports pre-trained models for **text classification, summarization, translation, question answering, and more**.

Key Features:

- BERT, GPT, RoBERTa, T5, etc.
- Easy-to-use pipelines for NLP tasks
- Supports PyTorch & TensorFlow

Example (Sentiment Analysis):

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")
result = classifier("I love studying Natural Language Processing!")
print(result)
```

Output:

```
[{'label': 'POSITIVE', 'score': 0.9998}]
```

6. Comparison of Libraries

Library	Best For	Pros	Cons
NLTK	Learning & research	Rich resources, flexible	Slower, not production-ready
SpaCy	Production & speed	Fast, accurate, pretrained models	Limited beginner tutorials
Gensim	Topic modeling & embeddings	Efficient with large corpora	Limited NLP functions beyond embeddings
TextBlob	Quick prototyping	Simple API, sentiment analysis	Not scalable for large datasets
Hugging Face Transformers	Deep learning & SOTA models	Powerful, pre-trained models	Requires more computing resources

MULTIPLE CHOICE QUESTIONS

 Which of the following tasks belongs to Natural Language Generation (NLG)?
a) Sentiment detection
b) Keyword extraction
c) Automatic report generation
d) Spam filtering
Ans: c
 Which NLP step converts raw text into machine-processable tokens?
a) Lemmatization
b) Parsing
c) Tokenization
d) POS tagging
Ans: c
 ChatGPT mainly performs which NLP function?
a) NLU
b) NLG
c) Tokenization
d) POS tagging
Ans: b
 Which is the first step in the NLP pipeline?
a) POS tagging
b) Lemmatization
c) Tokenization
d) Parsing
Ans: c
 Which library is primarily used for transformer-based NLP models?
a) NLTK
b) SpaCy
c) HuggingFace
d) TextBlob
Ans: c
 Which NLP technique may produce non-dictionary words?
a) Tokenization
b) Lemmatization
c) Stemming
d) POS tagging
Ans: c
 Which NLP task helps in understanding grammatical structure?
a) Tokenization
b) Parsing
c) Lemmatization

d) Stopword removal
Ans: b
 Google Translate is an example of:
a) Information retrieval
b) Machine translation
c) Chatbot
d) Text classification
Ans: b
 Which library is best suited for training word embeddings?
a) SpaCy
b) NLTK
c) Gensim
d) TextBlob
Ans: c
 Which of the following is NOT a preprocessing step?
a) Tokenization
b) Stopword removal
c) Parsing
d) Lemmatization
Ans: c
 NLP is a combination of Artificial Intelligence and:
a) Networking
b) Linguistics
c) DBMS
d) Software Engineering
Ans: b
 Keyword extraction belongs to:
a) NLG
b) NLU
c) Parsing
d) Classification
Ans: b
 Which library efficiently supports Named Entity Recognition?
a) Gensim
b) SpaCy
c) NLTK
d) OpenCV
Ans: b
 NLP mainly processes which type of data?
a) Numeric
b) Structured
c) Unstructured text

d) Binary
Ans: c
 Removing punctuation is a part of:
a) Parsing
b) Normalization
c) POS tagging
d) Classification
Ans: b
 Which application uses both NLU and NLG?
a) Calculator
b) Chatbot
c) Spreadsheet
d) Compiler
Ans: b
 POS tagging assigns:
a) Topics
b) Grammar roles
c) Sentiments
d) Keywords
Ans: b
 Which library is specifically designed for pretrained transformer models?
a) SpaCy
b) TextBlob
c) HuggingFace
d) NLTK
Ans: c
 The primary goal of NLP is:
a) Database optimization
b) Understanding human language
c) Network security
d) OS development
Ans: b
 Which is NOT an NLP application?
a) Machine translation
b) Image segmentation
c) Sentiment analysis
d) Chatbots
Ans: b
 Tokenization splits text into:
a) Labels
b) Words or sentences
c) Topics

d) Tags
Ans: b
 Future NLP research mainly focuses on:
a) Data storage
b) Bias and fairness
c) Hardware design
d) Networking
Ans: b
 Which is essential for chatbots?
a) Only NLU
b) Only NLG
c) Both NLU and NLG
d) Neither
Ans: c
 Which component converts structured data into human language?
a) NLU
b) NLG
c) NLP
d) POS tagging
Ans: b
 SpaCy is primarily written using:
a) Java
b) C++ and Python
c) PHP
d) Ruby
Ans: b

VERY SHORT ANSWER QUESTIONS

1. What is Artificial Intelligence?
- **AI is the simulation of human intelligence by machines.**
- **It enables learning, reasoning and decision-making.**

2. What is Natural Language Processing?
- **NLP is a branch of AI that processes human language.**
- **It helps computers understand and generate text.**

3. What is NLU?
- **NLU is Natural Language Understanding.**
- **It focuses on understanding the meaning of text.**

4. What is NLG?
- **NLG is Natural Language Generation.**
- **It generates human-like text from data.**

5. Mention any two applications of NLP.

- **Chatbots and virtual assistants.**
- **Machine translation and spam detection.**

6. What is Tokenization?

- **It splits text into words or sentences.**
- **It is the first step of NLP preprocessing.**

7. What is POS Tagging?

- **It assigns grammatical labels to words.**
- **It identifies nouns, verbs, adjectives, etc.**

8. What is Stemming?

- **It reduces words to their root form.**
- **It removes word suffixes.**

9. What is a Pre-trained Language Model?

- **It is a model trained on large text data.**
- **It improves NLP accuracy and speed.**

10. Name any two NLP libraries in Python.

- **NLTK and SpaCy.**
- **Gensim and TextBlob.**

LONG ANSWER QUESTIONS (PART A)

1. Explain Artificial Intelligence (AI) and its Types

Answer: Artificial Intelligence and Its Types

- Artificial Intelligence is the ability of machines to simulate human intelligence.
- AI enables machines to learn, reason and self-correct.
- It performs tasks that normally require human intelligence.
- Speech recognition and chatbots are AI applications.
- Narrow AI performs specific tasks.
- Siri and Google Search are Narrow AI examples.
- General AI performs multiple tasks like humans.
- Super AI exceeds human intelligence.
- General and Super AI are still theoretical.
- AI is the foundation for NLP systems.

2. Define NLP and Explain Its Applications

Answer: Natural Language Processing and Applications

- NLP is a subfield of AI that works with human language.
- It helps computers understand and generate text.
- NLP combines linguistics and computer science.
- Chatbots use NLP for customer service.
- Google Translate uses NLP for translation.
- Sentiment analysis uses NLP to detect emotions.
- Spam detection filters unwanted emails.
- Text summarization shortens long documents.

- Voice assistants use NLP for commands.
- NLP improves human–computer interaction.

3. Differentiate NLP, NLU and NLG

Answer: Difference Between NLP, NLU and NLG

- NLP processes human language in general.
- NLU understands the meaning of text.
- NLG generates human-like responses.
- NLP performs tokenization and POS tagging.
- NLU performs sentiment and intent detection.
- NLG creates chatbot replies and reports.
- NLP handles syntax and semantics.
- NLU focuses on context and meaning.
- NLG focuses on grammar and fluency.
- All three together build intelligent language systems.

4. Explain the Importance of NLP

Answer: Importance of Natural Language Processing

- NLP automates communication systems.
- Chatbots answer customer queries instantly.
- NLP analyzes large volumes of text.
- It extracts opinions from social media.
- NLP improves decision-making processes.
- It enhances voice assistant technologies.
- NLP supports multilingual communication.
- It helps in spam detection.
- It powers recommendation systems.
- NLP forms the backbone of AI applications.

5. Describe the Phases of NLP

Answer: Phases of Natural Language Processing

- Text preprocessing cleans raw text.
- Tokenization splits text into words.
- Stopword removal removes meaningless words.
- Stemming reduces words to root form.
- Language modeling predicts word sequences.
- Parsing identifies grammatical structure.
- POS tagging assigns grammar labels.
- NER identifies entities like names and places.
- Semantic analysis understands meaning.
- Application layer performs tasks like chatbots and translation.

6. Explain the History and Evolution of NLP

Answer: History and Evolution of NLP

- NLP started in the 1950s.
- Alan Turing proposed the Turing Test.
- Early NLP focused on machine translation.
- Rule-based systems used grammar rules.
- Statistical NLP used probabilities and corpora.
- HMM and N-gram models were introduced.
- Machine learning algorithms improved NLP.
- Naive Bayes and SVM were used.
- Deep learning introduced neural networks.
- Transformers like BERT and GPT revolutionized NLP.

7. Explain NLP Trends and Future

Answer: Trends and Future of NLP

- Pre-trained models reduce training effort.
- BERT and GPT improve accuracy.
- Chatbots are becoming more human-like.
- Multilingual NLP breaks language barriers.
- NLP analyses customer sentiments.
- Automated summarization is increasing.
- NLP is used in healthcare and education.
- Ethics and bias issues are emerging.
- Fair AI development is necessary.
- NLP will enable smarter automation.

8. Explain NLP Libraries in Python

Answer: NLP Libraries in Python

- NLTK is used for academic NLP learning.
- It supports tokenization and tagging.
- SpaCy is fast and production ready.
- Gensim supports topic modeling.
- Word2Vec is provided by Gensim.
- TextBlob is simple for sentiment analysis.
- Hugging Face supports deep learning models.
- BERT and GPT are available in it.
- Libraries simplify NLP development.
- Python is the most used NLP language.

9. Explain Text Preprocessing Techniques

Answer: Text Preprocessing in NLP

- Text preprocessing cleans raw data.
- Tokenization splits sentences into words.
- Stopword removal removes common words.

- Stemming reduces words to roots.
- Lemmatization finds dictionary base form.
- Lowercasing ensures uniform text.
- Cleaning removes punctuation.
- Preprocessing improves accuracy.
- It reduces noise in data.
- It is the first step of NLP pipeline.

10. Explain the Role of NLP in Modern AI Systems
Answer: Role of NLP in Modern AI

- NLP enables machine communication.
- Chatbots use NLP to respond.
- Translation systems use NLP.
- Search engines use NLP.
- Recommendation systems use NLP.
- Voice assistants use NLP.
- Spam filters use NLP.
- Social media monitoring uses NLP.
- NLP enhances automation.
- NLP supports intelligent AI systems.

LONG ANSWER QUESTIONS (PART B)

Case Study Title: Sentiment Analysis of Customer Reviews Using NLP
A leading e-commerce company wants to analyse millions of customer reviews to understand customer satisfaction and improve its product quality. They plan to use Natural Language Processing techniques to automatically classify reviews as positive, negative, or neutral.
Answer:

- Customer reviews are unstructured textual data.
- NLP helps convert text into machine-readable form.
- Tokenization splits reviews into words.
- Stop word removal cleans unnecessary words.
- Stemming and lemmatization normalize word forms.
- POS tagging identifies grammatical roles.
- Sentiment lexicons classify emotions.
- Machine learning models predict sentiment.
- Results help improve product quality.
- Customer satisfaction increases using NLP automation.

CASE STUDY – 2
Case Study Title: NLP-Based Chatbot for University Student Support
A university plans to develop a chatbot to answer student queries regarding admissions, fees, exam schedules, and results. The chatbot should provide quick and accurate responses without human intervention.

Answer:

- The chatbot uses NLP to understand student queries.
- Tokenization breaks queries into words.
- Text normalization removes noise.
- POS tagging identifies key terms.
- NLU detects student intent.
- NLG generates meaningful replies.
- Predefined datasets train the chatbot.
- NLP libraries like SpaCy and NLTK are used.
- The chatbot reduces staff workload.
- Students receive instant assistance.

UNIT II

LESSON PLAN

THEORY TOPICS NOTES

MULTIPLE CHOICE QUESTIONS

VERY SHORT ANSWER QUESTIONS

LONG ANSWER QUESTIONS (PART A)

LONG ANSWER QUESTIONS (PART B)

UNIT III

LESSON PLAN

THEORY TOPICS NOTES

MULTIPLE CHOICE QUESTIONS

VERY SHORT ANSWER QUESTIONS

LONG ANSWER QUESTIONS (PART A)

LONG ANSWER QUESTIONS (PART B)

UNIT IV

LESSON PLAN

THEORY TOPICS NOTES

MULTIPLE CHOICE QUESTIONS

VERY SHORT ANSWER QUESTIONS

LONG ANSWER QUESTIONS (PART A)

LONG ANSWER QUESTIONS (PART B)

UNIT V

LESSON PLAN

THEORY TOPICS NOTES

MULTIPLE CHOICE QUESTIONS

VERY SHORT ANSWER QUESTIONS

LONG ANSWER QUESTIONS (PART A)

LONG ANSWER QUESTIONS (PART B)